ACUGESIC INJECTION

DESCRIPTION:

ACUGESIC INJECTION is a clear, colourless, odourless solution.

COMPOSITION:

ACUGESIC INJECTION 50 MG (1ML AMP): 1ml solution for injection contains 50mg of Tramadol Hydrochloride. This is equivalent to one ampoule Acugesic 50mg.
ACUGESIC INJECTION 100 MG (2ML AMP): 2ml solution for injection contains 100mg of Tramadol Hydrochloride. This is equivalent to one ampoule Acugesic 100mg

PHARMACODYNAMICS:

The side effects of opioid therapy, manifested as respiratory depression and the development of dependence, which limit the application of potent analgesics, continue to impose on the therapist the difficult task of estimating the benefit-risk relation. Since it has not yet been possible to separate the therapeutical exploitable activity and the undesirable side effects of potent analgesics, it was assumed that the activity and side effects of potent analgesic were traceable to a common principle of action. Increasing knowledge suggests however that is possible to distinguish between the therapeutic effects and side effects.
Tramadol (INN) (±)–trans-2-(dimethylaminomethyl)-1-(m-methoxyphenyl)-cyclohexanol-hydrochloride is classified as a potent centrally active analgesic. Its structure is related to opiate alkaloids (eg. codeine) and to synthetic opiate analgesics.
Preclinical and clinical studies have shown that the analgesic action derives from the reactions of the drug with the specific opiod receptors of the pain processing system.
Animal experiments led to the assumption that tramadol hydrochloride is a substance in which the therapeutic effects and side effects can to a great extent be differentiated.
In tramadol HCl, with its favourable side effects/analgesic action relation, we now have available for the first time and opioids for the therapy of severe pain with a minimized abuse and dependence potential, and which in therapeutic doses usually induces no respiratory depression. Another advantage of Tramadol HCl is its absence of any euphoric component of action-a property, which in classical opioids tends to promote abuse and therapy contributes to the development of dependence.

PHARMACOKINETICS:

Tramadol is metabolized in man and animals mainly by N-and O-demethylation and subsequent conjugation. In contrast to animals much more intact tramadol is excreted with the urine in man; the cumulative renal excretion of Tramadol is about 10% (p.o) and 15% (i.v) with the exception of the metabolite O-demethyl-tramadol (M1), all other metabolite is pharmacologically inactive. Tramadol is rapidly distributed, giving a rapid onset of effect after intravenous administration. The biological half-life of the terminal phase in man (5.5h) is within a therapeutically favourable range. The main metabolites of tramadol are eliminated from the serum with comparable half-lives. Tramadol and its metabolites are excreted almost entirely via the kidneys.
On multiple administration of tramadol at six-hour intervals, the steady state serum concentrations are rapidly reached. The steady state Cmax value is about twice as high as after single administration.
There is no danger of undesirable accumulation of tramadol even with somewhat shorter intervals of administration. A significant increase in the incidence of side effects due to the accumulation of metabolites on repeated administration is unlikely since the main tramadol metabolites are eliminated in the terminal phase at the same rates as the parent drug.
The binding of tramadol to the proteins of human serum is not very marked. The fraction of protein-bound tramadol only amounts to about 4%. Thus, it does not exercise any major influence on the pharmacokinetics of the substances and can be discarded with respect to drug interactions.

INDICATION:

For the treatment of moderate to severe acute or chronic pain and in painful diagnostic or therapeutic measures.

RECOMMENDED DOSAGE:

The dosage should be adjusted to the intensity of the pain. Unless otherwise prescribed tramadol should be administered as follows: Single dose for adults and adolescents over 14 years of age:
i.v 50-100mg injected slowly or diluted in solution for infusion and infused.
i.m 50-100mg
In elderly (>75 years), dosage interval may be extended since elimination of tramadol may be prolonged.
In general, the daily adult dose should not exceed 400mg tramadol HCl. In impaired renal or hepatic function, it may be necessary to adjust the dose. Adults and adolescents (12 years and older)
Acugesic is not approved for use in patients below 12 years old. Paediatric population
The safety and efficacy of Acugesic has not been studied in the paediatric population. Therefore, use of Acugesic is not recommended in patients under 12 years of age.
Duration of Treatment:
During long-term treatment with Acugesic, the possibility of dependence cannot be entirely excluded. Therefore, the physician is to decide on the duration of treatment and whether the preparation is to be withdrawn temporarily.
Acugesic should not be given for longer than therapeutically necessary.

ROUTE OF ADMINISTRATION: For IM, IV or IV infusion

CONTRAINDICATIONS:

Known hypersensitivity to tramadol or any excipients.
Acute intoxication with alcohol, hynpotics, analgesics, opioids or psychotrophic drugs. Patients who are receiving MAO inhibitors or who have taken them within the last 14 days. Tramadol must not be used for narcotic withdrawal treatment.
Children younger than 18 years to treat pain after surgery to remove tonsils and/or adenoids.
Adolescents between 12 and 18 years who are obese or have conditions such as obstructive sleep apnea or severe lung disease, which may increase the risk of serious breathing problems.

WARNINGS & PRECAUTIONS:

Acugesic must be used with special care in patients with reduced level of consciousness of unclear origin, disorders of the respiratory centre or function, intracranial pressure, severe impairment of hepatic and renal function and in patients prone to convulsive disorders or in shock. Acugesic is not recommended as a substitute in opioid dependent patients. Although tramadol is an opiate-agonist, it cannot suppress opioid withdrawal symptoms. Withdrawal syndrome may occur if tramadol is discontinued abruptly. These symptoms may include: anxiety, sweating, insomnia, rigors, pain, nausea, tremors, diarrhea, upper respiratory symptoms, piloerection, and rarely hallucinations.
Acugesic should be used with care in patients with increased reactivity to opioids. Patients known to suffer from convulsions should be carefully monitored during treatment with Acugesic. Tramadol has a dependent potential. On long-term use, tolerance, psychic and physical dependence may develop. Therefore, in patients with a tendency towards drug abuse or dependence, treatment should only be carried out for short period under strict medical supervision.
Seizure Risk: Convulsions have been reported in patients receiving tramadol at the recommended dose levels. The risk may be increased when doses of tramadol exceed the recommended upper daily dose limit. In addition, tramadol may increase the seizure risk in patients taking other medication that lowers the seizure threshold (see Interactions). Patients with epilepsy or those susceptible to seizures should only be treated with tramadol if there are compelling circumstances.
Information for the patients: Acugesic is a potent drug for the relief of pain, eg in wound pain, fractures, severe nerve pain, tumour pain, heart attack. It should not be used for minor pain. The effect sets in quickly and last for some hours.
Effects on the ability to drive or operate machine: Even when administered according to instruction, Acugesic may affect the reaction ability of the patient to such an extent that his capacity to drive or operate machines may be impaired. This applies particularly in conjunction with alcohol.

<u>Paediatric population</u>
The safety and efficacy of Acugesic has not been studied in the paediatric population. Therefore, use of Acugesic is not recommended in patients under 12 years of age.

<u>Respiratory depression</u>
Administer Acugesic cautiously in patients at risk for respiratory depression, including patients with substantially decreased respiratory reserve, hypoxia, hypercapnia, or pre-existing respiratory depression, as in these patients, even therapeutic doses of Acugesic may decrease respiratory drive to the point of apnea. In these patients, alternative non-opioids analgesic should be considered. When large doses of tramadol are administered with anaesthetic medications or alcohol, respiratory depression may result. Respiratory depression should be treated as an overdose. If naloxone is to be administered, use cautiously because it may precipitate seizures.

<u>Cytochromes P450 (CYP) 2D6 Ultra-Rapid Metabolism</u>
Some individuals may be CYP2D6 ultra-rapid metabolisers. These individuals convert tramadol more rapidly than other people into its more potent opioid metabolites O-desmethyltramadol (M1). This rapid conversion could result in higher-than-expected opioid-like side effects including life-threatening respiratory depression. The prevalence of this CYP2D6 phenotype varies widely and has been estimated at 0.5 to 1% in Chinese, Japanese and Hispanics, 1 to 10% in Caucasians, 3% in African Americans, and 16-28% in North Africans, Ethiopians, and Arabs. Data are not available for other ethnic groups.

<u>Risks from Concomitant Use with Benzodiazepines</u>
Profound sedation, respiratory depression, coma, and death may result from the concomitant use of Acugesic with benzodiazepines. Observational studies have demonstrated that concomitant use of opioids and benzodiazepines increases the risk of drug-related mortality compared to use of opioids alone. Because of these risks, reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate.
If the decision is made to newly prescribe a benzodiazepine and an opioid together, prescribe the lowest effective dosages and minimum durations of concomitant use. If the decision is made to prescribe a benzodiazepine in a patient already receiving an opioid, prescribe a lower initial dose of the benzodiazepine than indicated in the absence of opioid, and titrate based on clinical response.
If the decision is made to prescribe an opioid in a patient already taking a benzodiazepine, prescribe a lower initial dose of the opioid, and titrate based on clinical response. Follow patients closely for signs and symptoms of respiratory depression and sedation. Advise both patients and caregivers about the risks of respiratory depression and sedation when Acugesic is used with benzodiazepines. Advise patients not to drive or operate heavy machinery until the effects of concomitant use of the benzodiazepine have been determined. Screen patients for risk of substance use disorders, including opioid abuse and misuse, and warn them of the risk for overdose and death associated with the use of benzodiazepines *(See Drug Interactions).*

<u>Serotonin Syndrome with Concomitant Use of Serotonergic Drugs</u>
Cases of serotonin syndrome, a potentially life-threatening condition, have been reported during concurrent use of Acugesic with serotonergic drugs (See Interactions with Other Medicaments). This may occur within the recommended dosage range.
Serotonin syndrome symptoms may include mental-status changes (e.g., agitation, hallucinations, coma), autonomic instability (e.g tachycardia, labile blood pressure, hyperthermia), neuromuscular aberrations (e.g., hyperreflexia, incoordination) and/or gastrointestinal symptoms (e.g nausea, vomiting, diarrhoea) and can be fatal (See Interactions with Other Medicaments). The onset of symptoms generally occurs within several hours to a few days of concomitant use, but may occur later than that. Discontinue Acugesic if serotonin syndrome is suspected.

<u>Adrenal Insufficiency</u>
Cases of adrenal insufficiency have been reported with opioid use, more often following greater than one month use. Presentation of adrenal insufficiency may include non-specific symptoms and signs including nausea, vomiting, decreased appetite, fatigue, weakness, dizziness, and low blood pressure. If adrenal insufficiency is suspected, confirm the diagnosis with diagnostic testing as soon as possible. If adrenal insufficiency is diagnosed, treat with physiologic replacement dosing of corticosteroids. Wean the patient off of the opioid to allow adrenal function to recover and continue corticosteroid treatment until adrenal function recovers. Other opioids may be tried as some cases reported use of a different opioid with recurrence of adrenal insufficiency. The information available does not identify any particular opioids as being more likely to be associated with adrenal insufficiency.

<u>Sexual Function/Reproduction</u>
Long term use of opioids may be associated with decreased sex hormone levels and symptoms such as low libido, erectile dysfunction, or infertility (See Postmarketing Experience)

INTERACTION WITH OTHER MEDICAMENTS:

Use with CNS Depressants:

Tramadol should be used with caution and in reduced dosage when administered to patients receiving CNS depressants such as alcohol, opioids, anaesthetic agents, phenothiazines, tranquillisers or sedative hypnotics.

The combination of tramadol with mixed opiate agonists/antagonist (e.g. buprenorphine, pentazocine) is not advisable because the analgesic effect of a pure agonist may be theoretically reduced in such circumstances.

Tramadol can induce convulsions and increase the potential for selective serotonin re-uptake inhibitors, tricyclic depressants, antipsychotics (Lithium) and other seizure threshold lowering drugs to cause convulsions.

Use with MAO inhibitors:

Tramadol should not be used in patients who are taking MAO inhibitors or who have taken them within the last fourteen days, as tramadol inhibits the uptake of noradrenaline and serotonin.

Use with Antiepileptic:

Concomitant administration of tramadol with carbamazepine causes a significant increase in tramadol metabolism, presumably through metabolic induction by carbamazepine. Patients receiving chronic carbamazepine doses of up to 800mg daily may require up to twice the recommended dose of tramadol.

Use with Cimetidine: Concomitant administration of tramadol with cimetidine does not result in clinically significant changes in tramadol pharmacokinetics. Therefore, no alteration of the tramadol dosage regimen is recommended.

Use with Benzodiazepines:

Due to additive pharmacologic effect, the concomitant use of opioids with benzodiazepines increases the risk of respiratory depression, profound sedation, coma and death.

The concomitant use of opioids and benzodiazepines increases the risk of respiratory depression because of actions at different receptor sites in the central nervous system that control respiration. Opioids interact primarily at μ-receptors, and benzodiazepines interact at GABAA sites. When opioids and benzodiazepines are combined, the potential for benzodiazepines to significantly worsen opioid-related respiratory depression exists.

Reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate (see Warning and Precautions). Limit dosage and duration of concomitant use of benzodiazepines and opioids, and follow patients closely for respiratory depression and sedation.

Use with Serotonergic Drugs:

The concomitant use of opioid with other drugs that affect the serotonergic neurotransmitter system has resulted in serotonin syndrome. If concomitant use is warranted, carefully observe the patient, particularly during treatment initiation and dose adjustment. Discontinue Acugesic if serotonin syndrome suspected. Examples of serotonergic drugs are selective serotonin reuptake inhibitors (SSRIs), serotonin and norepinephrine reuptake inhibitors (SNRIs) tricylic antidepressant (TCAs). triptants, 5-HT3 receptor antagonists, drugs that affect the serotonin neurotransmitter system (e.g., mirtazapine, trazodone, tramadol), monoamine oxidase (MAO) inhibitors (those intended to treat psychiatric disorders and also others, such as linezolid and intravenous methylene blue) (See Warning and Precautions)

USE IN PREGNANCY & LACTATION:

Use in Pregnancy: Tramadol has been shown to across the placenta. There are no adequate and well-controlled studies in pregnant women. Safe use in pregnancy has not been established. Acugesic is not recommended for pregnant women.

Use in Lactation: Approximately 0.1% of the maternal dose of tramadol is excreted in breast milk. In the immediate post-partum period, for maternal oral daily dosage up to 400 mg, this corresponds to a mean amount of tramadol ingested by breast-fed infants of 3% of the maternal weight-adjusted dosage. For this reason, tramadol should not be used during lactation or alternatively, breast-feeding should be discontinued during treatment with tramadol. Discontinuation of breast-feeding is generally not necessary following a single dose of tramadol.

SIDE EFFECTS:

Sweating, dizziness, nausea, vomiting, dry mouth, fatigue may occur after tramadol administration. It is also possible that the preparation may affect the cardiovascular system in rare cases. Undesirable effects occur particularly when the patient is under physical strain.

Respiratory depression (rare)

Post marketing Experience

Serotonin syndrome (See Warnings and Precautions) Adrenal insufficiency (See Warnings and Precautions)

Androgen deficiency: Cases of androgen deficiency have occurred with chronic use of opioids. Chronic use of opioids may influence the hypothalamic-pituitary- gonadal axis, leading to androgen deficiency that may manifest as low libido, impotence, erectile dysfunction, amenorrhea, or infertility. The causal role of opioids in the clinical syndrome of hypogonadism is unknown because the various medical, physical, lifestyle, and psychological stressors that may influence gonadal hormone levels have not been adequately controlled for in studies conducted to date. Patients presenting with symptoms of androgen deficiency should undergo laboratory evaluation.

Infertility: chronic use of opioids may cause reduced fertility in females and males of reproductive potential. It is not known whether these effects on fertility are reversible.

Special Note:

Respiratory depression has so far not been observed during treatment with tramadol. However, it cannot definitely be ruled out if the recommended dosage is considerably exceeded or on the concomitant administration of other centrally depressant drugs

SYMPTOMS AND TREATMENT OF OVERDOSE:

Symptoms:

In principles, on intoxication with tramadol, symptoms similar to those of other centrally acting analgesics (opioids) are to be expected, in particular miosis, vomiting, cardiovascular collapse, reduced level of consciousness up to coma, convulsions and respiratory depression up to respiratory arrest.

Treatment:

The general emergency measures to keep open the respiratory tract (aspiration), maintenance of respiration and circulation depending on the symptoms apply. An antidote for respiratory depression is naloxone. In animal studies, naloxone had no effect on convulsions. In such cases, diazepam should be given IV. Haemodialysis is not expected to be helpful because it removes only a small percentage of the administered dose.

INCOMPATIBILITIES:

Tramadol has proved to be incompatible with injection solutions of diclofenac, indomethacin, phenylbutazone, diazepam, flunitrazepam, glyceryl trinitrate.

STORAGE CONDITIONS:
Store below 30°C. Protect from light. Keep out of reach of children. *Jauhkan daripada kanak-kanak*.

SHELF LIFE:

Please refer to outer package.

PACK SIZE:
ACUGESIC INJECTION 50 MG (1ML AMP): Pack in 10's and 30's ampoules of 1 ml / box
ACUGESIC INJECTION 100 MG (2ML AMP): Pack in 10's and 30's ampoules of 2 ml / box

PRODUCT REGISTRATION HOLDER & MANUFACTURER:
DUOPHARMA (M) SDN BHD
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